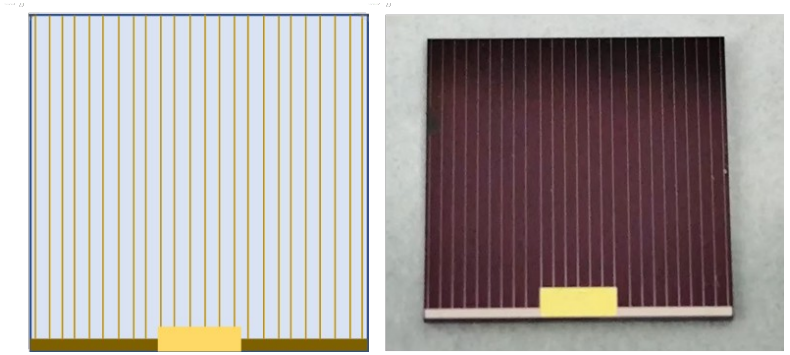
单结InGaAs：从结构制备到44.68%效率

1064 nm · 25±1 ℃ · 1 cm² | η = 44.68%　Voc = 0.765 V | 对应辐照度0.4 W/cm²

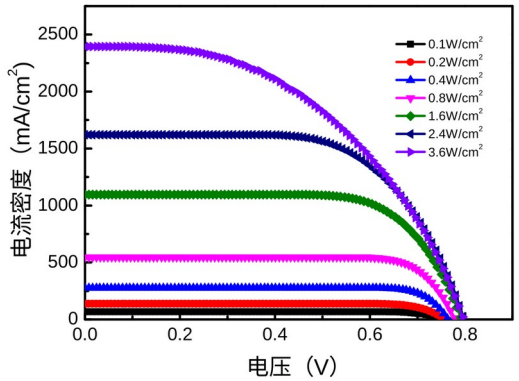
(a) 器件简化结构

GaInAs-Si Cap 层
GaInP-Si 窗口层
GaInAs-Si 发射区
GaInAs-Zn 基区
AlGaInAs-Zn 渐变缓冲层
AlGaAs/GaAs 隧穿结
GaAs-Si 缓冲层
N 型 GaAs 衬底

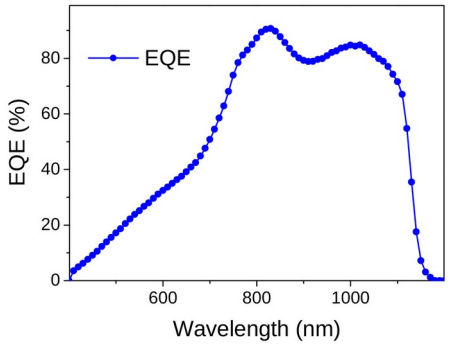
(b) 栅线与器件照片



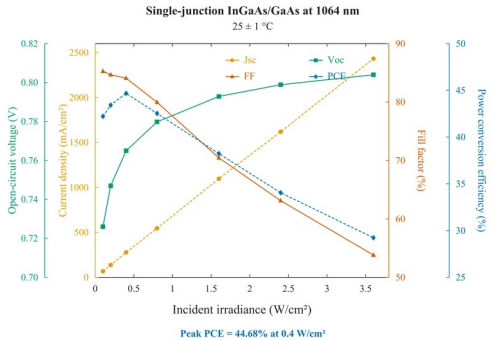
(c) I–V 曲线



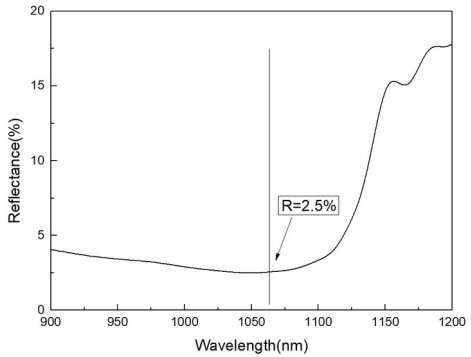
(d) EQE



(e) 辐照强度下的器件性能



(f) 反射率



减反与复合抑制协同；高辐照下受串联电阻限制。

来源：T1图3、图5、图6；TiO_x/Al_2O_3减反膜，R_{1064}=2.5%。